

10A Practice Exam (Question sheet)

Q1.

Simplify the following leaving in index notation

a) $(x^2 y^3)^4$

b) $\left(\frac{2xy^2}{xy}\right)^3$

Q2. Simplify the following, leaving answers with positive indices.

a) $\frac{x^4 y}{y^3} \times \frac{x^2 y^2}{x^3}$

b) $\left(\frac{x^2}{y^{-1}}\right)^{-2}$

Q3. Write each in index form

a) $\sqrt[5]{x^2}$

b) $(\sqrt{y})^3$

Q4. Evaluate the following

a) $8^{\frac{1}{3}}$

b)

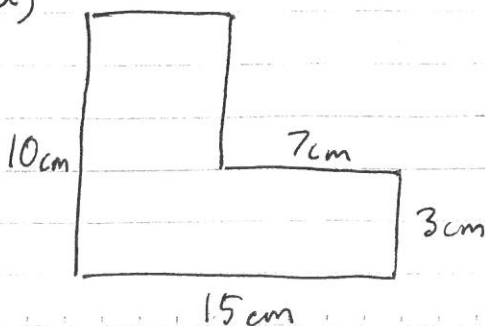
Q5. Simplify

a) $4\sqrt{7} + 9\sqrt{7}$

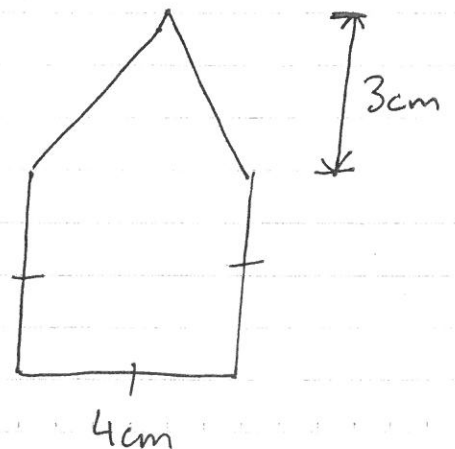
b) $5\sqrt{3} \times 3\sqrt{12}$

Q6 What is the area of this shape

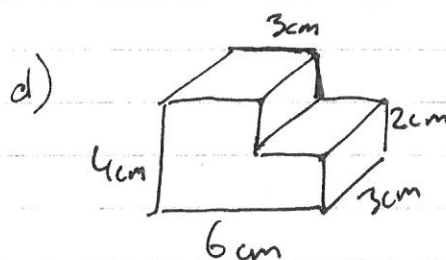
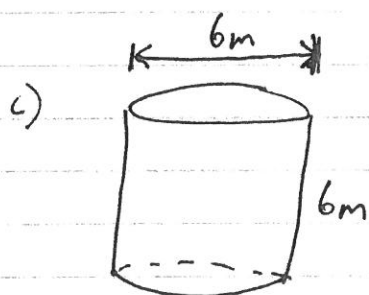
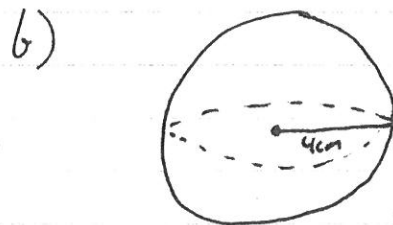
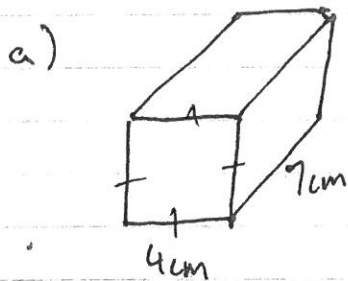
a)



b)



Q7. Find the volume and Surface area.



Q8 Solve the following simultaneous equations

a) Elimination method

$$\begin{aligned} x + y &= 5 \\ x - y &= 1 \end{aligned}$$

b) Substitution method

$$\begin{aligned} y &= 3x \\ 2x - y &= -6 \end{aligned}$$

c) graphically (draw graph, find point of intersection)

$$y = x + 3$$

$$y = 2x + 4$$

Q9. Expand the following

a) $(x+2)(x-2)$

b) $(2x+1)^2$

Q10. Factorise

a) $4x^2 + 18x$

b) $4xy + 8x^2 - 8x^3y$

Q11. Factorise the following

a) $x^2 + x + 4x + 4$

b) $u(u-2) + 7(u-2)$

c) $x^2 + 7x + 10$

d) $x^2 + 3x - 10$

e) $x^2 - 9$

f) $1 - y^2$

Q12 Solve the quadratics

a) $a^2 - 16 = 0$

b) $x^2 + 2x + 1 = 0$

Q13 Logs

Solve the following

a) $\log_5 625$

b) $\log_{10} 2 + \log_{10} 5$

Q14. Solve the inequality

a) $4x < 8$

b) $-2x \geq 16$

c) $x+5 < 21$

Q15. a) $\frac{x}{5} + \frac{2x}{5}$

b) $\frac{7x}{4} - \frac{3x}{4}$

c)

Q16 A box of crayons has 4 red, 2 black, 4 yellow, 2 blue, 3 green and 5 purple.

If a student takes 2 crayons at random

a) Draw a tree diagram

b) What's the probability of 2 red crayons

Q17.

A bag contains 2 yellow and 3 black marbles
Two are picked at random

a) Draw the sample space

b) $P(2 \text{ yellow})$

c) $P(\text{yellow and black})$

d) $P(\text{yellow given the first was black})$

Q18 50 people are asked whether they like cats or dogs.
40 people like dogs. 10 people like both cats and dogs
while 5 like neither.

Draw the venn diagram

How many people

a) only like dogs

b) like cats

c) only like cats